# Krishnasai Bairi

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## Professional Summary

Software Developer with **4 years of experience** building autonomous AI agents, enterprise automation platforms, and scalable Python backend services. Proven track record designing LLM-driven orchestration agents that cut exception resolution time by over 95% and eliminate 80%+ of manual operational work. Skilled in Python, REST APIs, distributed databases, agentic workflows, and workflow automation. Currently building production-ready Generative AI applications using FastAPI, Retrieval-Augmented Generation (RAG), semantic search, embeddings, vector databases, and modern backend architecture.

## Technical Skills

**Languages:** Python, JavaScript, SQL
**Backend:** FastAPI, REST APIs, Pydantic, Async Python, gRPC, Protocol Buffers
**Generative AI & Agents:** LLMs, AI Agents, Agentic Workflows, Retrieval-Augmented Generation (RAG), Semantic Search, Embeddings, Prompt Engineering, LangChain, LangGraph
**Databases:** Google Cloud Spanner, SQL, Vector Databases
**Tools & Practices:** Git, Docker, Blaze, Google Apps Script, Agile, CI/CD

## Professional Experience

**Software Engineer** — **Sep 2024 – Present**

- Architected and deployed an autonomous, LLM-driven orchestration agent to triage and resolve enterprise transaction validation failures across 5+ error classes, reducing Mean Time to Resolution (MTTR) by over **95% – from hours of manual work to under 5 minutes per transaction**.
- Automated more than **80%** of recurring operational exceptions in high-volume enterprise workflows using AI-driven orchestration and intelligent business rule engines, eliminating manual engineer intervention.
- Built Python services querying distributed relational databases (Google Cloud Spanner) and Protocol Buffers/gRPC interfaces to run automated integrity checks across transaction logs, resource mappings, and catalog data.
- Programmed dynamic escalation workflows that detect partner and vendor capabilities and automatically route updates through integration APIs and automated communications, streamlining cross-company operations.
- Designed a high-coverage automated test suite with mocked database fixtures under a modern build system, ensuring zero-regression production deployments.
- Developed Python data pipelines streaming real-time metrics from internal APIs into reporting dashboards, cutting manual tracking effort by **5–8 hours per week** and improving SLA visibility.
- Designed SLA and ticket-aging algorithms with priority-based and working-day calculations that auto-exclude weekends/holidays, achieving **100% calculation accuracy** and cutting report verification time by over **50%**.

**Associate Software Engineer** — **Aug 2022 – Aug 2024**

- Engineered a Google Apps Script alert-management system that monitors Gmail labels for critical system alerts, parses alert times/IDs/error details via regular expressions, and auto-creates/updates Buganizer issues with dynamic Markdown tables, logging all data to Google Sheets and saving **5–10 hours per week**.
- Built a data-driven error-reporting pipeline querying Plx (F1 engine) to cluster and categorize recurring errors, auto-creating or updating Buganizer issues and generating linked, per-error-group Google Spreadsheets with detailed transaction data, saving **8–12 hours per week** and accelerating root-cause analysis.
- Maintained a central tracking system integrating Google Sheets and Buganizer to eliminate redundant issue creation and manual spreadsheet updates.

## Projects

**Enterprise AI Knowledge Assistant** — Python | FastAPI | LangChain | LangGraph

- Built an enterprise AI assistant capable of answering organization-specific queries using Retrieval-Augmented Generation (RAG).
- Implemented semantic search using embeddings and vector databases to improve knowledge retrieval accuracy.
- Developed REST APIs using FastAPI with Pydantic models for request validation and structured response handling.
- Designed modular AI workflows using LangGraph and integrated Large Language Models for contextual conversations.
- Implemented prompt engineering techniques that improved response quality while reducing hallucinations.
- Built a scalable backend architecture following dependency injection and clean architecture principles.

## Education

**Bachelor of Technology, Computer Science and Engineering** — **2019 – 2022**
SR Engineering College, Warangal — CGPA: **8.8/10**

## Certifications

Associate Data Practitioner – Google